Hw

Sidneyville manufactures household and commercial furnishings. The Office Division produces two desks, rolltop and regular. Sidneyville constructs the desks in its plant outside Medford, Oregon, from a selection of woods. The woods are cut to a uniform thickness of 1 inch. For this reason, one measures the wood in units of square feet. One rolltop desk requires 10 square feet of pine, 4 square feet of cedar, and 15 square feet of maple. One regular desk requires 20 square feet of pine, 16 square feet of cedar, and 10 square feet of maple. The desks yield respectively \$115 and \$90 profit per sale. At the current time the firm has available 200 square feet of pine, 128 square feet of cedar, and 220 square feet of maple. The firm has backlogged orders for both desks and would like to produce the number of rolltop and regular desks that would maximize profit.

1. Formulate a mathematical programming model. How many of each should it produce?

Hw (Con.)

2. Perform sensitivity analysis. Based on the analysis, answer the following questions.

1. Sale manager has renegotiated the contract for regular desks and now expects to make a profit of $125 on each. He excitedly conveys this information to the firm's production manger, expecting that the optimal mix of roll top and regular desk will change as a result. Does it?
2. Suppose that the new contract also has a higher profit for the rolltop desks. If the new profit for the rolltop desks is $140, will this change the optimal solution?
3. A logging company has offered to sell an additional 50 square feet of maple for $5.00 per square foot. Accept the offer?
4. Assuming that the company purchases the 50 sq ft of maple, is the optimal solution affected?
5. The firm is considering a pine desk that would require 25 sq ft of pine and no other wood. What profit for pine desks would be required to make its production worthwhile, assuming current levels of resources and original profits on regular and rolltop desks?
6. During inspection, the quality department discovered that 50 sq ft of pine had water damage and could not be used. Will the product mix change?